

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)
[Introduction and Recap of Level I ▾](#)

< > Introduction and Recap of Level I



12.2 Applying Quantitative Techniques

Introduction and Recap of Level I

The Level I unit provided a comprehensive introduction to quantitative analysis in finance, emphasizing its accessibility and practical applications. The key learnings can be summarized as follows:

First, the investment process is a three-step process involving information, idea, and action. Information, both primary (raw data) and secondary (derived), feeds into the formation of an idea. This idea leads to action, which always involves some level of risk. Quantitative analysis plays a crucial role in this process by objectively measuring and comparing ideas, helping to overcome biases, and providing confidence in decision making. It serves as a feedback loop to either support or reject ideas, allowing investors to make informed decisions rather than acting on incomplete or biased information. This process applies to various actions in financial markets, from trading to advising clients or publishing market insights.

Second, quantitative analysis is presented as a disciplined approach to measuring ideas and strategies, rather than a complex field reserved for programmers or PhDs. It serves as a decision support tool, helping analysts evaluate the effectiveness of their strategies through rigorous testing and analysis. The unit stressed the importance of signal testing as the initial step in the quantitative process, followed by strategy testing and model building.

Third, the material covers essential prerequisites for conducting quantitative analysis, including proper universe selection, ensuring data quality, and understanding fundamental concepts such as rules, signals, strategies, and models. It also addresses critical considerations like survivorship bias in data, which can significantly impact the reliability of results. The unit provides guidance on using tools like Microsoft Excel for tracking signals and analyzing their effectiveness, focusing on key metrics such as probability of gain, average return, and standard deviation.

In this unit, we will step through the process of quantitatively testing an idea, and then building a strategy from it. By using this practical example, we will reinforce what was covered in Level I and introduce new concepts.

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12.2 Applying Quantitative Techniques



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Curriculum
Update Log



Sections



Search

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)

Defining a Quantitative Strategy ▾

< > Defining a Quantitative Strategy



12.2 Applying Quantitative Techniques

Defining a Quantitative Strategy

When we have spotted something in the market and we have come up with an idea of how it could be used in a trading system, the next thing we need to do is verify if this is an idea worth pursuing. While we showed in Level I how we can manually signal test an idea, it is much faster to take advantage of a tool that has been designed for this purpose. Quant-based tools can give us results in seconds, allowing us to focus on deciding whether the idea is worth pursuing.

Most quantitative tools will handle the management of universes and data, including the mitigation of common issues like survivorship bias. We barely need to think about them when we know the tools are taking care of these issues. You could achieve the same results using languages like Python or R, as long as you are aware of the issues and able to mitigate them.

The Full Quantitative Process

- Define your risk limits
- Select the universe
- Select the data time frame
- Define your rules
- Visualize the rule
- Signal testing
- Define your exit rules
- Define stops
- Strategy testing
- Analyze the results
- Optimization

We will take the process from Level I and build upon it. In this unit, we will look at these steps in greater detail.

Define Your Risk Limits

We need to know what our risk limits are. As we get older, our ability to withstand risk commonly diminishes. Hopefully, our investment balance is large enough that we no longer need to take on aggressive risk to achieve the level of returns we need. Knowing our risk limits allows us to reject ideas that over-trade or trade when volatility is too high.

Knowing our risk tolerance before we begin allows us to filter out strategies that are not appropriate for us or our clients.

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12.2 Applying Quantitative Techniques

Select Our Universe and Data Time Frame

We covered the selection of a universe and the importance of quality data in Level I, but we did not discuss the data time frame. In technical analysis, we often teach that our indicators are fractal, in that we can apply our indicators to data on any and every time frame.

If we were using intraday data, we would have many more signals, and the subsequent trades would be shorter in duration. If we were using weekly data, we would get fewer signals, and the trades would be longer.

Several factors may influence our choice of which time frame to use.

It should be noted that while we may use weekly or monthly time frames for our signals, accurate testing tools will calculate the statistics and monitor the stops on a daily or intraday basis. This is to imitate real-life portfolio management where our technical analysis may be calculated on weekly charts, but we monitor the portfolio daily.

Constraint	Time Frame to Use
Security is highly leveraged	Shorter
Account preservation is most important	Longer
Trading costs are high	Longer
Account size is small	Shorter

Signal-to-Noise Ratio

In audio applications, we transmit signals down a wire (e.g., from a microphone to a speaker). From the first telegraphs in the 1800s, engineers have been fighting the interference of noise on electrical circuits. Their role was to ensure the signal strength was high enough for the signal to be received and the noise was filtered out. The good thing was that typically the noise was a constant, and increasing the power of the signal allowed the signal to get through.

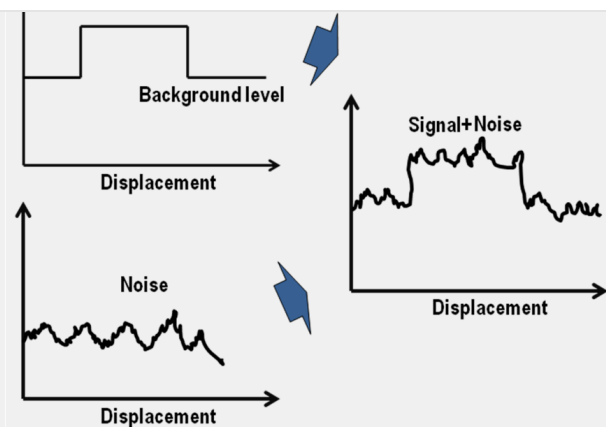


FIGURE 12.2.1 Combination of signal and noise

Source: [ESR](#)

It is helpful to think of market trends in the same way. In every time frame, there are trends we can take advantage of. Our technical analysis is designed to capture when these trends change so we can participate in the trend. The issue is that we must consider noise in the markets. News stories, rumors, opinions, etc., can all impact a security's trend in the short term, but it usually does not change the long-term trend.

The shorter our time frame, the more susceptible our signals are to capturing noise trends rather than a tradable trend.

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Curriculum
Update Log



Sections



Search



Curriculum
Update Log



Sections



Search

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)
[Defining Your Rules](#)
[<](#)
[>](#)

Defining Your Rules



12.2 Applying Quantitative Techniques

Defining Your Rules

Define the Entry Rule

No matter which tool you use to help you, the codification of your rule is up to you. Each vendor has a library of examples on how you can code up rules, and there are countless examples online if you are building your own system. In some cases, vendors have trained artificial intelligence (A.I.) models that can be your assistant. The ability to use A.I. to help code your ideas opens up quantitative analysis to analysts who were previously unable to codify their rules.

Rule Types

There are three types of rules we work with. Each type is used for a different purpose, and if they are mixed up, the tests will give erroneous results. The three types are:

Trigger Rules

Trigger rules are true only on the date when the condition is satisfied. This is when a signal should be generated. For example, the crossing of price over a 200-period moving average is a discrete event that is only true on the day of the cross.

```
Close() CrossesAbove MA(BARS=200)
```

Filter Rules

Filter rules are true for many days at a time. They are used to ignore signals that may have satisfied the trigger rule but should be rejected because of another factor. For example, we may find our signal only performs well when the benchmark index has closed higher. If this were the case, we would add a filter to only accept signals when the index closed higher.

```
Index() IsUp
```

Value Rules

The trigger and filter rules only return a true or false result. Value rules return a numeric value and are used for ranking. When we have many signals triggered on the same day, a decision needs to be made about which signals are selected first. A value rule is included in the strategy to set an execution order. For example, we may favor securities with lower volatility. We can rank the list by setting the Average True Range as our value rule and then sort them in ascending order with the values moving lowest to highest.

```
ATR(BARS=14)
```

RRG Idea

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12.2 Applying Quantitative Techniques

visualizing relative strength relationships. We all know buying into an equity after its big move is too late (anyone buy Nvidia late in 2024?). It is better to buy an equity as it is just about to rise in value. Can the RRG give us a clue about when a security has just begun rising?

Our idea could be that when an equity is in the “Lagging” quadrant and crosses into the “Improving” quadrant, it is a good time to buy that equity. Here is the trigger rule for that, using the S&P 500® (SPX) as the benchmark:

```
JDKRS (INDEX=SPX:WI) .Quadrant ChangeTo 3
```

This simple RRG rule is used to trigger when the arrow enters the “Improving” quadrant, or quadrant 3.

The first time we test a new idea, we do not want to add any filter rules. We will create those once we see the results of the first test.

Figure 12.2.2 shows the trigger about to happen on an RRG chart. The arrow is just about to leave the “Lagging” quadrant and move into the “Improving” quadrant, or quadrant 3.

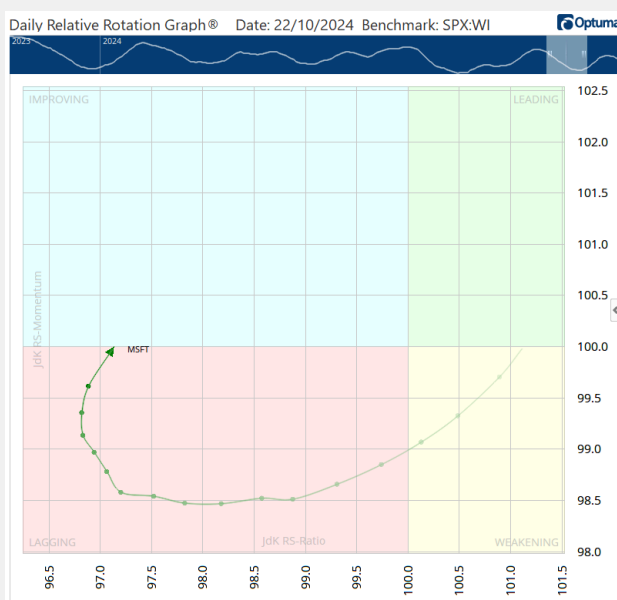


FIGURE 12.2.2 Relative Rotation Graph showing the path of Microsoft (MSFT) relative to S&P 500 index

Visualizing Our Rule

Before we test our rule, it is important to visualize it. This allows us to confirm our rule is capturing the signal we are after. In this chart, the bars are colored to match the location of the equity on the RRG (e.g., green when in the “Leading” quadrant, red when in “Lagging,” etc.).

The green arrows are set to the same script we will use in our test.

Curriculum
Update Log

Sections

Search



FIGURE 12.2.3 Microsoft bar chart with RRG quadrant colors. Green arrows display our signal.

We see on the chart our rule is correctly capturing the idea, since the bars turn blue as the security enters the “Improving” quadrant, but when we look at it on the chart, we see some examples where it would not have performed very well. It is possible for us to think about a filter at this stage, but for this example, we will just run the signal test and come back to that later.

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 Curriculum Update Log

 Sections

 Search



Curriculum
Update Log



Sections



Search

Topics

Table of Contents

Confidence Levels

Notes

Bookmarks

Signal Testing

Signal Testing



12.2 Applying Quantitative Techniques

Signal Testing

In the Level I unit on quantitative analysis, we discussed the theory of signal testing and performed a manual signal test using Excel. In this unit, we will jump straight into using Optuma and run our RRG test.

Test Parameters

- Use S&P 500 stocks, accounting for survivorship bias.
- Use daily data.
- Test range is 20 years from 2001 to 2021.
- For each signal, we measure 21 days after the signal.
- We also measure five days before the signal.

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Results

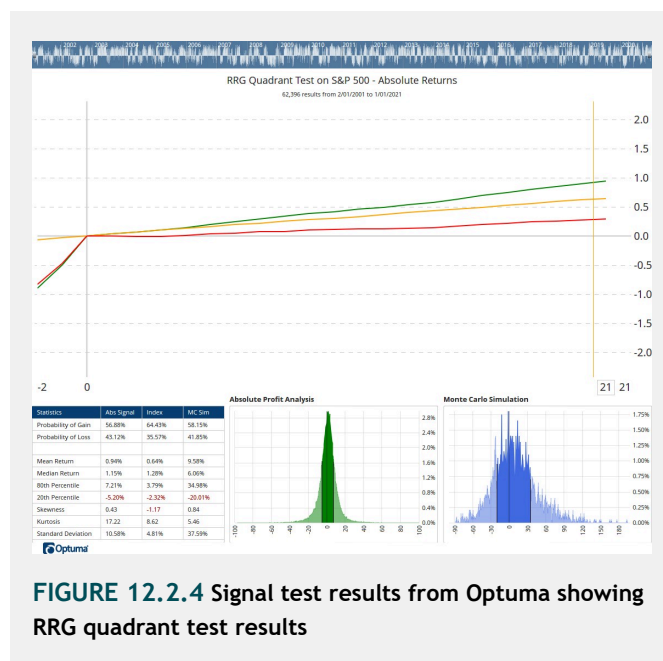


FIGURE 12.2.4 Signal test results from Optuma showing RRG quadrant test results

Recall from Level I that the yellow plot is the average returns of the Index. The red line shows the average outperformance or difference between the green line (our idea) and the yellow line (the benchmark).

The key measures we are looking at are as follows:

1. **Mean return of 0.94%**
Annualized, this would be around 11.9%.
2. **Probability of gain is 56.9%**
This is getting close to our minimum target of 60%, but we should do better.

more confidence we can have to hit our mean value. Nevertheless, this value is small enough.

4. Over 62,000 signals

This is a high number, and more signals than we could ever trade. We need many signals so we can find opportunities in the market when we are ready to trade, so this is more than enough. If our rule is too unique and results in too few signals, then we may need to wait weeks to trade. It is also a function of how long we would want to hold the trade. If we were looking for a short mean reversion trade, then lots of signals are good. Too few signals (fewer than 1,000) is something to avoid unless it captures a unique, high-probability opportunity.

5. The signal's average return (green line) is virtually straight

This is positive, as it means we are capturing an idea that is persistent for the month. Note that before "0" (to the left of the vertical line), the green plot rises sharply. On an RRG, a security moving into the "Improving" quadrant means its relative trend is improving, and the fact that the signal's return is also rising suggests that the absolute trend is improving, too.

6. Profit analysis is leptokurtic

This gives us confidence the results can be repeated. It should be noted that in a normal distribution, we would see high leptokurtic as an issue, since it suggests fat tails. (See the unit titled "Introduction to Probability" in Level I for a full explanation on measuring a distribution's kurtosis, or the degree to which returns show up in the tails of a distribution.)

7. Yellow benchmark plot is below the green plot on the main graph

This tells us our signal performs better than the benchmark returns (the red plot is the relative outperformance value).

8. The history bar at the top of the results (see Figure 12.2.5 for details) shows thousands of white lines

Each line represents a signal, and if it is drawn from the center up, it tells us that signal ended in a profit. On examination, profits and losses are grouped together. This suggests we can find a filter to turn off the rule when the losses are occurring.

9. Monte Carlo plot is not great

This is produced by taking 10 random signals and compounding their returns. We repeat that 2,000 times and generate a frequency distribution of those returns. We want to see this with positive skew. (See the unit titled "Introduction to Probability" in Level I for a full explanation on measuring a distribution's skew, or length of the tails of a distribution.)



FIGURE 12.2.5 History bar showing every time we get a signal and how it returned

risk. At this point, we could say our hypothesis that this idea would add value has been rejected.

Quantitative analysis is not “do one test and then reject or accept.” Rather, it is an iterative process where we make slight adjustments to our idea based on each test and our knowledge as technical analysts.

Signal Test #2

When we are looking at a relative strength indicator, like RRG, we need to consider that a security can be outperforming the benchmark on a relative basis, while one or both are still trending down.

One filter we can add is to only take the RRG signal when the benchmark (the S&P 500 index) is above its three-day moving average (a signal line). We could use a MACD for this, too.

Our new script would look like this:

```
// Store the change to RRG Improving Trigger
in T1
T1 = JDKRS(INDEX=SPX:WI).Quadrant ChangeTo
3;

// Get the data for the S&P 500 index
S1 = GETDATA(CODE=SPX:WI);

// Calculate the Moving Average of S1
M1 = MA(S1, BARS=3, CALC=Close);

// Store the condition that S1 is greater
than M1
F1 = S1 > M1;

// return a Trigger when both T1 and F1 are
true
T1 and F1
```

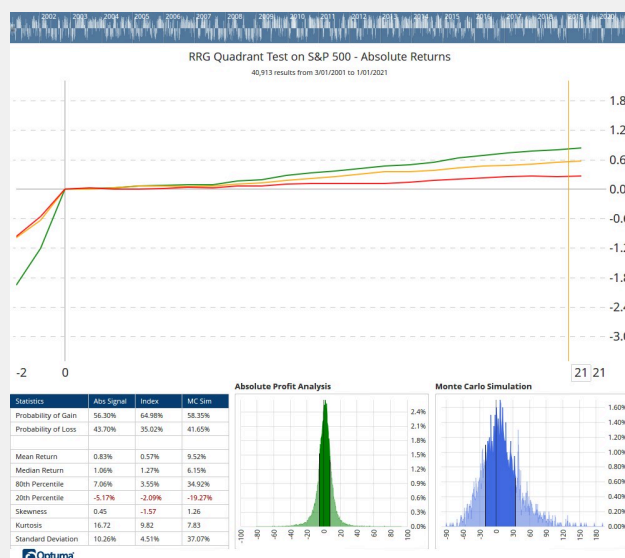


FIGURE 12.2.6 Signal test results from Optuma showing RRG quadrant test results when the benchmark is above its three-day average

	Test #1		
Mean Return	0.94%	0.83%	Less return
Probability of Gain	56.9%	56.3%	
Standard Deviation	10.6%	10.3%	
# of Signals	62,396	40,913	
Green Plot		Green plot did not outperform the benchmark for the first six days.	If the performance after day six was amazing, we could use that information to delay our signal by six days.

Summary

This result rejects the idea that the trend of the benchmark has an influence on our signals, or at least with a three-period MA as the filter.

Signal Test #3

It would take too long to show all the adjustments we could make to improve our signal. Instead, we will jump straight to the final iteration on this idea. Looking once more at the RRG chart with a single security on it (this is 100 days of observations for Microsoft), we can see multiple rotations. Some are close to the center of the chart and others are further away.

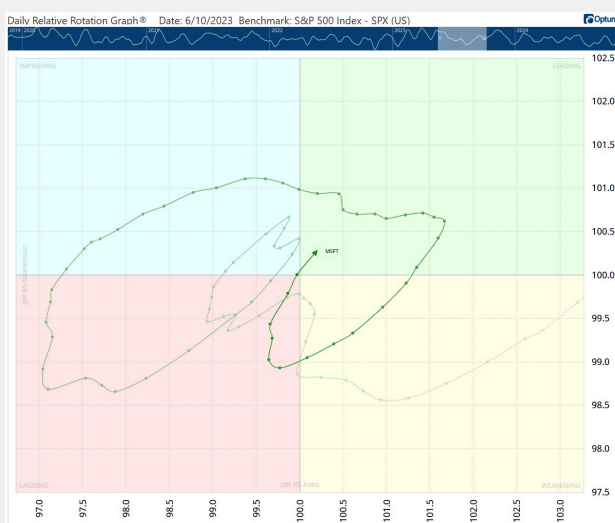


FIGURE 12.2.7 Relative Rotation Graph showing the path of Microsoft (MSFT) relative to S&P 500 index

```
// Store the RRG calculation in R1
R1 = JDKRS(INDEX=SPX:WI);

// Filter 1 is when the security is heading
between 300 degrees and 20 degrees
F1 = (R1.Heading > 300) or (R1.Heading <
20);

// Filter 2 is when the security is at least
3 units away from the center of the RRG
F2 = R1.Distance > 3;

// Trigger is security entering Improving
quadrant
T1 = R1.Quadrant ChangeTo 3;

//Signal when all 3 conditions are true
T1 and F1 and F2
```

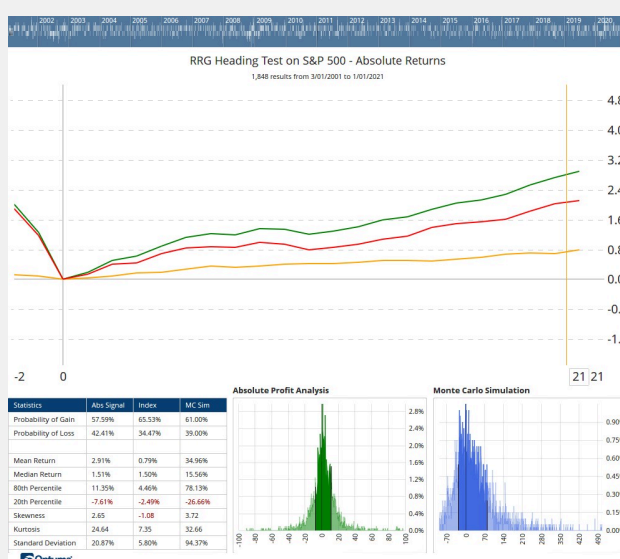


FIGURE 12.2.8 Signal test results from Optuma showing RRG quadrant test results when MSFT enters the “Improving” quadrant, but the heading is between 20 and 300 degrees and it is at least three units away from the center of the RRG.

The key measures for this final test are:

- 1. Mean return of 2.91%**
This is a good return (41% annualized).
- 2. Probability of gain at 57.6%**
We would accept this since the mean return is good.
- 3. Standard deviation is 20.9%**
For signal dispersion, this would be moderate. It is important to note that standard deviation includes negative deviations and positive deviations.
- 4. 1,848 signals**
This could be too small and not give us enough signals to work with.
- 5. The green signal average return plot significantly outperformed the yellow benchmark with an initial burst**

7. Good distribution of signals over the test period with no clumps of losing signals

8. Monte Carlo plot is positively skewed

	Signal Test #1	Signal Test #2	Signal Test #3
Mean Return	0.94%	0.83%	2.91%
Probability of Gain	56.9%	56.3%	57.6%
Standard Deviation	10.6%	10.3%	20.9%
# of Signals	62,396	40,913	1,848

Summary

This is something we can work with. We will continue to build a strategy around this in the next sections.

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Sections



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Sections



Search

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)
[Defining Exits](#)
[<](#)
[>](#)

Defining Exits



12.2 Applying Quantitative Techniques

Defining Exits

Define the Exit Rules

Signal testing exit rules are not as simple as entry rules. Exit signals are only applicable when they occur after we have entered a trade. When we are in a trade, we have a specific condition that the exit needs to be tested under.

There are different ways we can deal with this:

1. Add our entry condition into the exit rule and only trigger if we are within a time window from the entry. Scripting this can be difficult, but it would allow us to signal test our exit idea.
2. Use a hybrid rebalance model, where we only close the trade when new signals are triggered. We will be investigating rebalancing in Level III.
3. Use a simple exit that captures a change in trend. From a Dow Theory perspective, our entry signals are trying to capture trends. Ideally, a new trend is soon confirmed, and we profit as the market climbs higher. An ideal exit would be when the trend turns against the trade. We can use trend indicators to help.

Swing Charts as Trend Indicators

A powerful group of trend indicators is swing charts. They only have two states, up or down, so they are great for identifying trends. With all swing charts, there are objective rules to define when the swing changes direction. This makes them perfect for our use.

In Figure 12.2.9, we have a Gann swing chart on Microsoft Corp. The rules require three higher bars (higher top and higher bottom) to change the swing direction to “up.”

There are other rules that Gann introduced where the trend is not considered to be changed until a previous low (or high) has been breached.¹ Figure 12.2.9 is colored according to these rules, with uptrend being green and downtrend being red. This change of trend indication is a calculation that can be used with any swing chart.

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12.2 Applying Quantitative Techniques

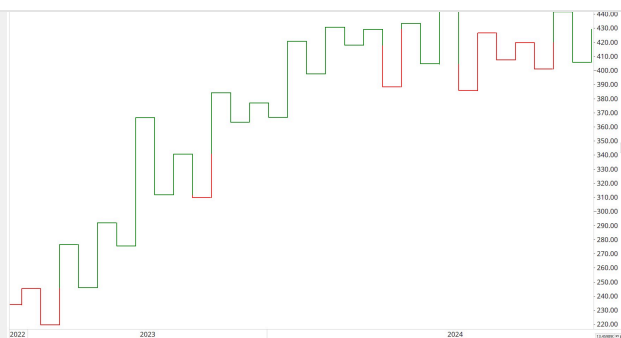


FIGURE 12.2.9 Microsoft Corp. Gann swing chart with trend colors

Another swing type is a volatility swing where the change only happens when the price of the security has moved more than a fixed number of Average True Ranges (ATRs) from the previous top or bottom.

Exiting on a Change in Trend

For our exit, we will use a volatility swing set to three ATRs and use Gann's rules for defining the change in trend to only happen when a previous swing-low has been breached.

Figure 12.2.10 shows four RRG entry signals on HP Inc. with green arrows.



FIGURE 12.2.10 Hewlett Packard Inc. with green arrows showing RRG signals

The second and third signals were not so good; hopefully our exit can allow us to capture the up moves and avoid big losses.

There is a function that makes this script simple to use. Here is our exit:

```
SwingTrendDown(VolatilitySwings(ATRS = 3.00))
```

We will add this to the chart as red arrows. We want to do a visual analysis of the first red arrow that occurs after our green signal, which we can see in Figure 12.2.11.



FIGURE 12.2.11 Hewlett Packard Inc. with entry signals (green) and exit signals (red)

Visually, this looks like a good exit signal. Signal 2 made a loss and signal 3 made a small profit, but signals 1 and 4 made good returns. This is something we can work with.

Stops

Before we build our strategy test, we need to address stops. Stops are critically important, and no matter how good the results from tests are, you should always have stops in place.

The first time we run our test, however, we will always do so without stops. Stops affect the result of the test, and it is only once we have completed our first test that we have an idea of the types of stops we should use.

We will come back to stops after we run our strategy test.

¹ Gann, William D. *Truth of the Stock Tape*. New York: Financial Guardian Publishing Co., 1923.

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 Curriculum Update Log

 Sections

 Search



CMT Curriculum Level II ▾



Notifications



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My Account



Curriculum
Update Log



Sections



Search

Topics

Table of Contents

Confidence Levels

Notes

Bookmarks

Strategy Testing

< >

Strategy Testing

12.2 Applying Quantitative Techniques

Strategy Testing

Position Size

There is one final step before we can run our strategy, and that is to define the position size. There are several different calculations we can use, from simple allocation of capital to volatility-based calculations. We will cover several position-sizing options in detail in Level III.

In this test, we will use a simple equally weighted allocation. That is, we determine the maximum number of positions we want to allow at one time, and then divide up our available capital accordingly. A good place to start is 4 to 5%. Having 20 to 25 securities in a portfolio is generally accepted as well diversified.

Once we run our first strategy test, we will get an idea about how our capital is being used, and if we need to make any adjustments.

Test the Strategy

The strategy testing process, like the signal testing process, is iterative. It is rare that we will have a perfect strategy right from the start. Sometimes we need to adjust our position size; other times we need to add stops. The first test will give us clues on what we need to do.

When we run our tests, the first thing we should do is examine the plots. Within seconds, we can see if there are any issues we need to deal with.

Equity Plot

The most common display we look at is the equity plot (cumulative returns plot) as shown in Figure 12.2.12. It shows the total value of our portfolio (including cash and holdings). Ideally, we want to see this increase over time.

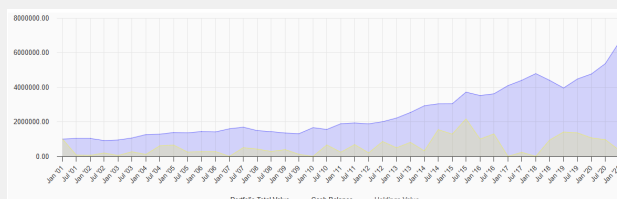


FIGURE 12.2.12 Strategy test equity plot showing the total value of the portfolio over the test and the cash balance

Cash Plot

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Unit

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close as possible. The first two things we want to check are the number of trades and the cash balance plot on our equity plot.

In our initial strategy test, we did not use up the cash. If we did, there would only be spikes of yellow on the plot in Figure 12.2.12 after a trade was exited, and it would quickly disappear as the funds were allocated to a new position.

The large periods of growing yellow tell us we are exiting positions, but not finding new trades to take. The main reason for this is we have a strict filter stopping us from taking a trade when the S&P 500 index is falling. We do put almost all the cash to work again in 2017, so there is no issue with the system's ability to use the cash when required.

Looking at the statistics (not displayed), the total number of trades is reported as 907. Remember from our signal test we had 1,800 signals. If the signals are well distributed through the test period, we can increase the number of holdings in the portfolio (by reducing the position size) to capture more signals.

Rolling Returns

The equity plot in Figure 12.2.12 is seducing. We might see the starting number and the ending number and start preparing for a long vacation. It can be difficult to consider all of the time that passed between the start and the end – it was 20 years. More realistic plots are the rolling monthly, quarterly, and yearly returns. For every day in our test, we look back by those three periods and calculate our return.

Figure 12.2.13 shows the rolling yearly returns. Each point on the graph shows the return from 12 months prior. Ideally, this would be a straight horizontal line as that would mean consistent returns throughout the life of the test.

Many presenters will show returns plots like Figure 12.2.12 because the hyperbolic growth is exciting and makes their results look wonderful. Remember that consistent yearly returns are more important.

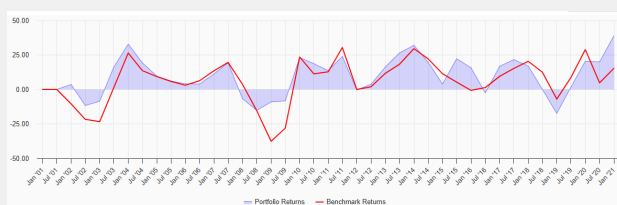


FIGURE 12.2.13 Rolling 12-month returns of the strategy

Drawdowns (Underwater Plot)

It is inevitable there will be periods when our strategy loses money. When the market falls, our portfolio will fall with it. The drawdown (or underwater) plot measures any retracement from the peaks in the equity plot in Figure 12.2.14. What is immediately obvious is that our portfolio performed better than the general market. Our S&P 500 index trend filter helped with this.

recovered.

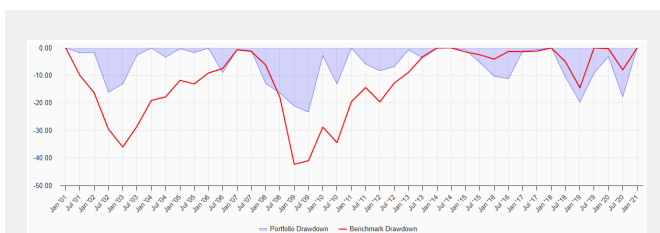


FIGURE 12.2.14 Drawdown plot for the RRG strategy

Income

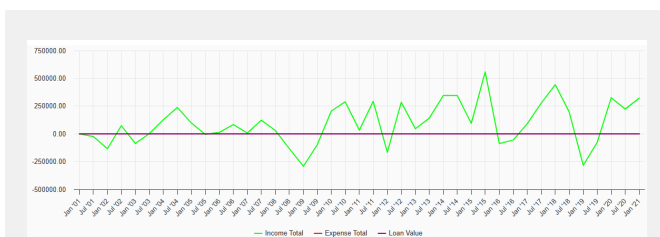


FIGURE 12.2.15 Income and expenses plot

The financials plot in Figure 12.2.15 shows when the strategy generated income. We did not set any brokerage fees on this initial test, so the expenses are flatlined.

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Curriculum Update Log

Sections

Search



Curriculum
Update Log



Sections



Search

^ High

= Medium

v Low

Continue >>

Unit

12.2 Applying Quantitative Techniques

12.2 Applying Quantitative Techniques

Comparing Strategies

There is an infinite number of combinations of rules and settings we can apply to our strategy. The goal of quantitative analysis is to test our idea and allow us to compare it to other ideas, or other iterations of our idea, to give us the confidence to act upon it.

There are several metrics we can use to measure the performance of our strategy. All have their pros and cons, so it is best to compare a number of these measures when evaluating strategies.

In this section, we will explore the most common measures used in quantitative finance. For each, we will highlight how they are calculated and how they are used. We will also group each of the measures into accounting, trade, capital asset pricing model (CAPM), and performance measures.

This list is not exhaustive, and more measures will be introduced in Level III.

Annualized Returns and Annualized Volatility

Many of the performance measures below are derived from annualized returns and annualized volatility. The most common way these are calculated are from the monthly returns of the strategy (or portfolio). The first step is to measure each month's return from the start of our test to the end. In a 20-year test, there will be 240 monthly values.

From these monthly returns, the average return is calculated by a simple arithmetic average. The standard deviation of the returns is also calculated. It is important to remember we work with the monthly returns in decimal form (e.g., 0.132) and not percent (e.g., 13.2%) when performing these calculations.

We then annualize these values using the two formulae below:

$$\text{Annualized Return} = \{[(1 + \text{MonthlyReturn})^{12}] - 1\} * 100$$

$$\text{Annualized Volatility} = \text{MonthlyVolatility} * \sqrt{12}$$

Accounting Measures

Accounting measures are focused on the overall financial performance of the strategy.

Total Return

Total return is the overall return on the starting capital.

$$TR = \left(\frac{\text{EndingValue} - \text{StartingValue}}{\text{StartingValue}} \right) * 100$$

CAGR

Maximum Drawdown

Maximum drawdown is the numerical value from the drawdown plot in Figure 12.2.14. It is a measurement of the largest peak-to-trough decline, useful for understanding worst-case scenarios in investment performance.

Drawdown Recovery Time

For all the drawdowns, what was the greatest number of days for our total equity to recover? These two drawdown numbers are important because they give us an idea of the negative periods we may need to endure.

Benchmark Returns

All the accounting measures, annualized return and annualized volatility, are repeated for the returns in the benchmark. These values show where our strategy outperformed the benchmark, but they are also used in some of the performance measures below.

Risk-Adjusted Returns

Our strategy has higher returns than the benchmark, but that usually comes with higher volatility. The CAPM tells us we should get more return as we expose ourselves to more risk. The only way we can be sure we are outperforming the benchmark is to adjust the risk in our results to scale the returns to the benchmark before checking the values.

What would our return be if we had the same volatility as the benchmark? We would reject our idea if the risk-adjusted returns are less than the benchmark's return.

$$RiskAdjustedReturns = \left(\frac{R_p}{\sigma_p} \right) * \sigma_b$$

Where R_p is the return of the portfolio, σ_p is the standard deviation of the returns of the portfolio, and σ_b is the standard deviation of the benchmark.

$$M^2$$

M^2 , also known as the Modigliani risk-adjusted performance, measures the return of the portfolio relative to the return of the benchmark. M^2 is similar to the risk-adjusted return, but factors the risk-free rate into the calculation.

$$M^2 = \left(\frac{R_p - R_f}{\sigma_p} * \sigma_b \right) + R_f$$

Where R_p is the return of the portfolio, R_f is the risk-free rate, σ_p is the standard deviation of the returns, and σ_b is the standard deviation of the benchmark.

You may recognize the first part of this formula is the same as the Sharpe ratio. M^2 can be calculated by multiplying the Sharpe ratio by the benchmark's standard deviation and then adding back the risk-free rate.

CAPM Measures

The capital asset pricing model (CAPM) is a fundamental concept in finance that helps investors understand the

Rate Your Confidence

Unit

12.2 Applying Quantitative Techniques

Comparing Strategies



Rate Your Confidence

Alpha

Alpha represents the excess return of our strategy relative to the benchmark. It measures the strategy's ability to beat the market. A positive alpha indicates outperformance, and a negative alpha indicates underperformance.

Beta

Beta measures the volatility or systematic risk of our strategy in comparison to the benchmark. A beta of 1 indicates the strategy moves in sync with the market, while a beta greater than 1 suggests higher volatility than the market. A beta less than 1 indicates lower volatility. Beta helps us understand how sensitive our strategy is to systemic risk.

There are two drivers that affect beta:

1. Correlation
2. Volatility

To have a beta of 1, the strategy would have high correlation to the benchmark and similar volatility.

For low beta, there are two options:

1. Low correlation, similar volatility
2. Low volatility, same correlation

For high beta, the volatility is the main driver. If we have perfect correlation and high volatility, then we will have high beta.

$$r^2$$

Alpha and beta are derived from a regression calculation. When we imagine a regression scatter plot with a line of best fit applied to the chart, r^2 is how tightly the observations are clustered around the line. The closer the value is to 1, the tighter the observations are to the line and the more confidence we can have in the alpha and beta values. Technically, r^2 is called the "coefficient of determination."

Trade Measures

This group of measurements consists of simple counts or averages from all the trades recorded in the strategy.

Number of Trades

The number of trades lets us know if the strategy has more trades than we can handle. There is no point implementing a strategy that trades a hundred times a day if you cannot automate your trade execution.

Gross Profit

Gross profit is the sum of all profitable trades in the portfolio. Gross profit ignores any other sources of income (dividends, interest, etc.), but does include brokerage, since that directly affects the profit from the trade.

Gross Loss

Unit

12.2 Applying Quantitative Techniques

Comparing Strategies



from the gross profit.

Rate Your Confidence

If our strategy produces low net profit, there are two possible reasons:

1. The gross profit is low. We would need to focus on our rules.
2. The gross profit is high but so is the gross loss. The gross loss would be eroding our net profit. We would need to investigate if we can use better filters to avoid trading when the largest losses happen and investigate if stops can reduce the losses without impacting the gross profit.

Profit Factor

The profit factor is a simple ratio that compares the total gross profit from the strategy with the total gross loss. A value of 1 would tell us our losses are equal to our profit. We would expect a number greater than 1.

$$ProfitFactor = \frac{GrossProfit}{GrossLoss}$$

Average Trades per Month

The number of trades divided by the number of months in the test. It allows us to get a feel for how much trading would be required to follow this strategy.

Average Profit

Average profit per trade. This measure has limited value, as the trade sizes usually grow as profits are reinvested.

Average Return

Average return per trade measured in percentages. Average return is a better measure than average profit, as average return is more consistent over the life of the test.

Average Trade Length

The average time a trade was held.

Longest Winning Trade Length

If this is many years, then a single trade may be contributing too much to the strategy's results. It could be a scenario that is unlikely to happen again and is therefore misleading.

Longest Losing Trade Length

Same as above, except it could have too much negative influence on our results.

Win Percent

The percentage of trades that ended in profit.

Winning Trades

The number of trades that ended in profit.

Average Win

The average size of all profitable trades.

Average Win Return

The average return from all trades that ended in a profit.

Unit

12.2 Applying Quantitative
Techniques

Comparing Strategies



The number of trades that ended in loss.

Rate Your Confidence

Average Loss

The average size of all losing trades.

Average Loss Return

The average return from all trades that ended in a loss.

Expected Value

Expected value is the profit (or loss) you can expect to make from the next trade. The value considers the probability of a win multiplied by the average win size, less the probability of a loss multiplied by the average loss size.

$$\text{ExpectedValue} = (\text{WinPercent} * \text{AvgWin}) - (\text{LossPercent} * \text{AvgLoss})$$

note: *AvgLoss* is an absolute value, meaning the number used in the equation is a positive value.

Largest Profit

The single largest profit recorded in the test. It would be concerning if the largest trade profit was close to our overall profit. We should not have a single trade that impacts our final results by more than a few percent. That is, if the final profit was \$2,000,000 and after 20 years, the largest profit was \$200,000, we would have to investigate. The probability of repeating this result in the future would be low.

Best Return

The single largest profitable return. This is not always the same as the largest profit.

Largest Loss

The single largest loss recorded in the test. If this is too large, it may drag down the strategy's results unfairly.

Worst Loss

The single largest losing return.

Max Adverse Excursion (MAE)

While a trade may end in a profit, it is often the case the trade was losing money for a period. When we first enter a trade, we are in deficit because of the brokerage. If the trade goes against us and we lose money, the deficit will grow. The deficit is called the "adverse excursion," but only if the trade recovers and ends in profit. The MAE is the greatest adverse excursion for all the trades in the strategy that ended in a profit.

MAE helps us set our initial stop levels, as setting stops beyond the MAE will likely not result in prematurely exiting trades that will ultimately be profitable.

Max Favorable Excursion (MFE)

This is similar to MAE, except it measures the peak in profit before the trade closed. Again, we only consider trades that ended in a profit, but it is often the case that we do not exit a trade until we get confirmation the trend has changed. MFE tells us the maximum return we could have made.

Unit

12.2 Applying Quantitative Techniques



Comparing Strategies



the trade's risk.

Risk is defined as the value we are risking by taking the trade. When we are using a stop, the risk is the trade price minus the stop price. When we are not using a stop, the whole trade is what we are risking. The risk is equal to the entry price.

$$RMultiple = \frac{Profit}{Entry\ Price - Stop}$$

This formula presents profit in terms of the risk the trade was exposed to. It punishes profits that came from elevated levels of risk. It is a form of risk adjusting the profit of each trade.

R-Multiple Examples

Trade at \$1,000. Stop at \$800 (risking \$200):

- Exit at \$1,200. R-Multiple would be 1R
- Exit at \$1,400. R-Multiple would be 2R
- Stopped out at \$800. R-Multiple would be -1R

Trade at \$1,000. No stop:

- Exit at \$1,200. R-Multiple would be 0.2R

Average R-Multiple

The values for R-Multiple are calculated for each trade, and then the average of all the R-Multiples is calculated as the average R-Multiple. The goal is to have an R-Multiple > 1R. This would imply that our average profit is greater than our average risk.

Performance Measures

This next group of measurements are calculations we use to compare strategies. They are derived from the values we just discussed.

All the measures consist of a return value on the numerator and a tracking error on the denominator. Typically, the tracking error is a standard deviation. If the deviations are small, the tracking error will be small, and it will have little impact on the performance measure. If there is a lot of variation, the tracking error will be large and will reduce the value of the performance measure.

Sharpe Ratio

The Sharpe ratio is a key performance measure that helps us gauge how well our strategy is performing relative to the risk we are taking. Essentially, it compares the excess return of our strategy (that is, the return above the risk-free rate) to its volatility. A higher Sharpe ratio suggests we are getting more return for each unit of risk we are taking on. It is a handy tool for comparing strategies, but there are some issues with the Sharpe ratio, which we will discuss in Level III. We need to consider Sharpe alongside other measures to get a comprehensive view of our strategy's performance.

The formula for the Sharpe ratio is straightforward:

$$Sharpe = \frac{R_p - R_f}{\sigma_p}$$

Rate Your Confidence

Unit

12.2 Applying Quantitative
Techniques

Comparing Strategies



The Information ratio is a key performance metric that measures a portfolio manager's ability to generate excess returns relative to a benchmark. It compares the portfolio's active return (the difference between the portfolio returns and the benchmark returns) to the active risk (tracking error). A higher Information ratio indicates that a manager has been more consistent in delivering outperformance relative to the benchmark. This makes it particularly useful for evaluating active management strategies and comparing different portfolio managers.

The formula for the Information ratio is:

$$\text{Information Ratio} = \frac{\text{Average}(\text{ExcessReturns})}{\text{StdDev}(\text{ExcessReturns})}$$

Where ExcessReturns is $R_p - R_b$, or the difference between the portfolio returns and the benchmark returns. All values should be annualized for consistency.

Sortino Ratio

The Sortino ratio is similar to the Sharpe ratio. While Sharpe looks at all volatility, Sortino only cares about downside volatility. It's like saying, "We're okay with upside surprises, but how much are we getting punished for the downside?" This makes it particularly useful for strategies that might have asymmetric return distributions. A higher Sortino ratio suggests better risk-adjusted performance, specifically in terms of managing downside risk.

The formula for the Sortino ratio is:

$$\text{Sortino} = \frac{R_p - R_f}{\sigma_d}$$

Where R_p is the return of the portfolio, R_f is the risk-free rate, and σ_d is the standard deviation of the downside returns. As always, ensure all values are annualized for consistency.

Treynor Ratio

The Treynor ratio, also known as the reward-to-volatility ratio, is another tool in our quantitative toolbox. It is similar to the Sharpe ratio but with a twist. Instead of using total risk (standard deviation), it uses systematic risk (beta). This makes it particularly useful when we're dealing with well-diversified portfolios. The Treynor ratio tells us how much excess return we are getting for each unit of market risk we are taking on. A higher Treynor ratio suggests better risk-adjusted performance. It is important to note that Treynor cannot be used with negative beta.

The formula for the Treynor ratio is:

$$\text{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

Where R_p is the return of the portfolio, R_f is the risk-free rate, and β_p is the beta of the portfolio. As with the Sharpe ratio, it's crucial that all values are annualized for consistency.

Calmar Ratio

The Calmar ratio is particularly useful when we are dealing with strategies that might experience significant drawdowns.

Rate Your Confidence

Unit

12.2 Applying Quantitative
Techniques

Comparing Strategies



suggests better risk-adjusted performance, indicating the strategy is delivering good returns relative to its maximum drawdown. However, it is important to note the Calmar ratio can be sensitive to the time period chosen, especially if it includes periods of exceptional market performance or stress.

The formula for the Calmar ratio is:

$$\text{Calmar Ratio} = \frac{\text{Annualized Returns}}{\text{Maximum Drawdown}}$$

While the maximum drawdown is the largest peak-to-trough decline over the same period, remember that both these values should be expressed as positive percentages for the calculation.

SQN - System Quality Number

The SQN is defined by Van K. Tharp in his book *Definitive Guide to Position Trading* to categorize trading systems using the R-Multiple value we calculated earlier. One criticism is that it favors systems with large numbers of trades and narrow volatility.

$$SQN = \left(\frac{\text{Average}(RMultiple)}{\text{StdDev}(RMultiple)} \right) * \sqrt{NumTrades}$$

The SQN value should be deemed reliable when the number of trades ≥ 30

When determining the quality of various systems, Tharp uses the following as a guide:

- 1.6 - 1.9 Below average
- 2.0 - 2.4 Average
- 2.5 - 2.9 Good
- 3.0 - 5.0 Excellent
- 5.1 - 6.9 Superb
- ≥ 7.0 - Holy grail

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Unit

12.2 Applying Quantitative Techniques

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 Curriculum
Update Log

 Sections

 Search

< >

Comparing Strategies



Rate Your Confidence

Unit
12.2 Applying Quantitative
Techniques

Rate Your Confidence

Unit

12.2 Applying Quantitative
Techniques

 Curriculum
Update Log

 Sections

 Search

< >

Comparing Strategies



Rate Your Confidence

Unit
12.2 Applying Quantitative
Techniques



Curriculum
Update Log



Sections



Search



Comparing Strategies



Rate Your Confidence

Unit

12.2 Applying Quantitative
Techniques



Curriculum
Update Log



Sections



Search



Comparing Strategies



Rate Your Confidence

Unit

12.2 Applying Quantitative
Techniques

Curriculum Update Log

Sections

Search

< >

Comparing Strategies



Rate Your Confidence

Unit
12.2 Applying Quantitative
Techniques



Curriculum
Update Log



Sections



Search



Comparing Strategies



Rate Your Confidence

Unit

12.2 Applying Quantitative
Techniques

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)
[Strategy Results](#)

< >

Strategy Results



12.2 Applying Quantitative Techniques

Strategy Results

Let us have a look at the results for our strategy test.²

Measure	Value	Comment
<u>Portfolio Performance</u>		
Total Return	459%	This looks amazing until you remember this is over 20 years.
Annual Return	11.7%	The average monthly return that is annualized. This value would be okay if the annual volatility was lower.
Annual Volatility	22.5%	This is too high for the amount of return.
CAGR	9.0%	
Risk-Adjusted Returns	8.6	
M^2	6.85	
Sharpe Ratio	0.54	The value reflects the low returns vs. the high volatility.
Sortino Ratio	2.05	Sortino only considers the downside standard deviation. The large difference between the Sharpe ratio and the Sortino ratio highlights that we must have some big winners that are increasing the overall volatility.
Treynor Ratio	8.2	Beta close to 1 means Treynor is close to the excess returns.
Calmar Ratio	0.14	The large drawdown has impacted this ratio.
Maximum Drawdown	64.3%	

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Continue >>

Unit

12.2 Applying Quantitative Techniques

Curriculum
Update Log

Sections

Search

MDD Days	1,144	This is how many days it took to recover the lost value in the portfolio.
Average R-Multiple Value	0.1	This is a low number because we had no stops on our test. Remember the R value is the return divided by the amount risked. Each position was fully risked.
SQN	5.2	Even though our test was not great, we still get an "Excellent" result here. This highlights why we cannot rely on a single value for comparison. The high value is due to low variability in the R value for each trade and the high number of trades.
<u>CAPM</u>		
Alpha	0.01	This highlights that we have low excess returns over the benchmark. This is a very low number and is part of the reason that the results were not as good as we hoped.
Beta	1.1	With the beta over 1, it suggests our portfolio is sensitive to volatility in the benchmark.
r^2	0.66	A value of 1 is perfect, so this means we have moderate to strong confidence in our alpha and beta values.
<u>Benchmark Performance</u>		
Total Return	192%	
CAGR	5.5%	
Annual Return	4.4%	
Annual Volatility	16.61%	
Sharpe Ratio	0.25	
Maximum Drawdown (MDD)	56.8%	
MDD Days	2,088	

 Curriculum Update Log

 Sections

 Search

Trade		
Performance		
Trades	760	
Gross Profit	\$8,475,150	The gains are impressive.
Gross Loss	- \$3,891,236	We gave back too much. We would want to find a way to limit these losses if we were to take this strategy to market.
Net Profit	\$4,583,914	
Profit Factor	2.18	
Average Profit	\$16,272	
Average Return	8.5%	
Max Adverse Excursion	-89.8%	This is a high value. We need to investigate using a stop.
Max Favorable Excursion	104%	This indicates the exit rule is waiting too long before exiting. We need to investigate different rules.
Winning Trades	399	
Win Percent	52.5%	
Average Win	\$21,241	
Losing Trades	360	
Loss Percent	47.4%	
Average Loss	-\$10,809	
Trading Expectancy	\$6,028.06	
Max Profit	\$480,176	This came from FCX:US on March 30, 2020. We should investigate this trade and see if it makes sense. We could find an issue we need to consider in our strategy.
Max Return	312%	This came from FCX:US on March 30, 2020. While this is the same trade as max profit, it is not always the case.
Max Loss	-\$184,760	This came from NBL:US on October 4, 2020.
Min Return	-93.37%	This came from BSC:US on February 27, 2009.



² Note: With the numbers provided in this unit, it is not possible for the candidate to calculate all of the equations in this table. They would need the results of the strategy test and all of the individual trades. The focus here is on interpreting the values.

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Curriculum
Update Log



Sections



Search



Curriculum
Update Log



Sections



Search



Curriculum
Update Log



Sections



Search

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)

Stops ▾

< > Stops



12.2 Applying Quantitative Techniques

Stops

Without using stops, we are risking too much in every trade. The R-Multiple measure of 0.1 highlights the relationship between the stop and the risk we are accepting for each trade. When we trade, we will inevitably be wrong a large portion of the time. Therefore, it is important we set stops that can release the capital from a bad trade as soon as possible. We then reassess what the market is doing and look for new opportunities.

From a Dow Theory perspective, we know we should only trade in the direction of the primary trend. When we trade, we have the belief the security will continue in the direction of the primary trend. There will be many secondary trend pullbacks, but if a pullback becomes a new primary trend in the opposite direction, we need to stop out of the position.

Types of Stops

There are two main types of stops: fixed stops and trailing stops. Within each type, several calculations can be applied.

Fixed Stops

Fixed stops are set when the trade is entered and are not changed. Their purpose is to exit the trade if the level is breached. It is essentially a safety net that will stop the loss.

- **Price**
Price stops are the most common. They are simply a level where the trade will be exited. Traders will often look at prior support and resistance levels and position their price stops at those levels. The theory is that a breach of the previous levels would signify a change in trend.
- **Percent**
Percent stops are similar to price stops, except the level is defined by the maximum loss the trader is willing to accept. For example, if we are willing to lose 20% on our trade at \$100, we would set our stop at \$80.
- **Time**
Time stops exit a trade after a period of time. This stops our trades from languishing and hoarding capital that could be used on new opportunities when they are not triggering the exit rule.
If signal testing showed that our entry rule performed amazingly for the first 10 days after the signal before flattening and providing minimal returns, then we may want to have a time stop after 10 days so we can release the capital and enter a new signal that has a

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Continue >>

Unit

12.2 Applying Quantitative Techniques

Trailing Stops

Trailing stops follow the market as the trade becomes more profitable. They typically only move in a positive direction, and will trigger if the price of the security changes trend and crosses the last peak level. They are sometimes called a “ratchet stop” because they can only move in one direction.

Trailing stops are more than a disaster stop, and they are sometimes used as the exit rule for a strategy.

- **Percent**

Percent was listed as a fixed stop above, but it can also be used as a trailing stop. The stop remains a percentage interval behind the price. If the price moves in a positive direction, the stop level is updated to the security’s price minus the percentage. The stop never decreases in value. Using our example from above, when we enter at \$100 with a 20% trailing stop, the initial stop is at \$80. As the price grows to \$140, the new stop level is \$112.

- **Volatility**

A fixed percentage may not be suitable for all securities our strategy trades. A 5% stop on a low volatility stock may be too great, and a tighter stop may be more suitable. Similarly, on a high volatility stock, the 5% stop may be too tight and be triggered too soon. The solution to this is a volatility stop that uses one of the volatility calculations to define the stop level. The most common is the two-ATR stop, where the level is set to twice the size of the ATR for that security.

- **Equity**

This works the same way as the percent trailing stop above, except that an equity stop tracks the overall value of the strategy and monitors the drawdown in equity. If the strategy’s equity falls below the threshold, all positions are exited. The equity stop acts like a reset where the losing trades are exited and capital is released for new opportunities.

Targets

Targets are similar to trailing percentage stops, except they are used to exit a trade once a profit level has been reached. From the signal testing, we saw very few signals generated more than 20% profit. Setting a target at 20% to exit at least half the trade could generate better results.

RRG Strategy Results with Stops

By adding a 30% trailing stop to our RRG strategy, we can see the impact the performance measures.

Measure	Original Value	New Value	Comment
<u>Portfolio Performance</u>			
Total Return	459%	561%	Good improvement


Curriculum Update Log


Sections


Search

			in overall performance.
Annual Return	11.7%	11.5%	
Annual Volatility	22.5%	17.5%	One of the benefits of stops is their ability to reduce the volatility in the strategy.
CAGR	9.0%	9.9%	
Risk-Adjusted Returns	8.6	10.9	
M^2	6.85	8.51	
Sharpe Ratio	0.54	0.5	Sharpe is only slightly lower since the volatility is reduced by a similar proportion.
Sortino Ratio	2.05	2.85	
Treynor Ratio	8.2	10.5	
Calmar Ratio	0.14	0.25	The smaller drawdown benefits this result.
Maximum Drawdown (MDD)	64.3%	40%	Higher than 30, this happens when a security gaps past the stop level.
MDD Days	1,144	901	
Average R-Multiple Value	0.1	0.1	
SQN	5.2	4.4	
<u>CAPM</u>			
Alpha	0.01	0.01	
Beta	1.1	0.84	
r^2	0.66	0.64	
<u>Benchmark Performance</u>			No changes, as we would


Curriculum Update Log


Sections


Search

			expect.
Total Return	192%	192%	
CAGR	5.5%	5.5%	
Annual Return	4.4%	4.4%	
Annual Volatility	16.61%	16.61%	
Sharpe Ratio	0.25	0.25	
Maximum Drawdown (MDD)	56.8%	56.8%	
MDD Days	2,088	2,088	
<u>Trade Performance</u>			
Trades	760	907	The increase in the number of trades is because the funds from the stopped-out trades are released to be used in new opportunities.
Gross Profit	\$8,475,150	\$10,520,248	
Gross Loss	- \$3,891,236	-\$4,913,581	
Net Profit	\$4,583,914	\$5,606,667	
Profit Factor	2.18	2.14	
Average Profit	\$16,272	\$17,016	
Average Return	8.5%	7.3%	
Max Adverse Excursion	-89.8%	-24.3%	The significant drop here is due to our stop exiting trades below 30.
Max Favorable Excursion	104%	56.5%	
Winning Trades	399	434	


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Sections


Search

Win Percent	52.5%	47.9%	Our percentage of winning trades has gone down below 50%. This is okay if the size of our winners is bigger than our losers.
Average Win	\$21,241	\$24,240	
Losing Trades	360	472	
Loss Percent	47.4%	52%	
Average Loss	-\$10,809	-\$10,410	
Trading Expectancy	\$6,028.06	\$6,197.76	
Max Profit	\$480,176	\$518,616	
Max Return	312%	336.86%	
Max Loss	-\$184,760	-\$126,839	
Min Return	-93.37%	-38.02%	We can see here the stop has made a difference. The extra 8% will be due to price gapping past the stop level.

Stops have improved the performance of this strategy, but there will be more we can do.

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Sections



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Update Log



Sections



Search



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Sections



Search

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)
[Optimization](#) ▾

< > Optimization



12.2 Applying Quantitative Techniques

Optimization

It can be tempting after a mediocre testing result to optimize the parameters in the strategy and find the right combination of parameters that yields the best return.

If we were using a 50-period simple moving average (SMA) in a strategy, we could set up an optimization on that length parameter of 50, starting at 25 and ending at 75 with a step of 5.

This would give us the following parameters for the SMA: 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75.

The software would run the strategy 11 times, and each time use the next parameter for the SMA. The goal of the optimization process is to identify the SMA parameter that results in the highest return in the strategy.

This optimization example would be in one dimension, as it is only one parameter. The number of tests that need to be run rises exponentially as dimensions are added to the test. The table below shows the total number of tests required if each parameter had 10 steps and the time to run a test was only one minute.

Parameters	Number of Tests	Time to Complete
1	10	10 min
2	100	1 hr 40 min
3	1,000	16 hr 40 min
4	10,000	7 days
5	100,000	70 days

Data Mining

Data mining, in the context of quantitative analysis, refers to the practice of extensively searching through data to find patterns or relationships that may not be genuinely predictive or meaningful. This can lead to false conclusions or overfitted models.

According to Fabozzi, Focardi, and Kolm in their 2010 book *Quantitative Equity Investing: Techniques and Strategies*, data mining "refers to the practice of searching through large amounts of data to find statistically significant but possibly spurious patterns."³ They emphasize this can result in strategies

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Unit

12.2 Applying Quantitative Techniques

The key issue with running thousands of tests and selecting the set of parameters that yielded the best result is that we have “fit” the parameters to the data we are using for our test.

Bailey et al. in their 2014 paper “Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance” published in *Notices of the American Mathematical Society*, warn that “the more we search, the greater the probability of finding something that is not really there.”⁴

Overfitting

It is crucial to distinguish between legitimate data analysis and problematic data mining. Our goal is not to select parameters that yield the best return in the past but a set of parameters that yields good results no matter when they are used or on what universe of data.

An analogy would be to imagine we were creating a mask from strips of wet plaster. If we let the plaster hang loosely over our face and let it set, we would fail to fit the mask to our exact facial features. That mask could be worn by almost anyone. If, however, we hung the plaster and molded it to every contour of our face, we would have created a great mask that only fits us. No one else could use it since it was overfitted. This is what we do when we use optimization to select the best set of parameters. The middle ground is what should be aiming for, a strategy that alerts us to a change in trend without being over-optimized.

When we overfit the strategy through optimization, it is rare the results will be repeated when the strategy is traded live.

Robustness

Optimization should not be used without a measure of robustness. For example, if a 10-period SMA yields the best return, then we would expect a nine-period SMA and an 11-period SMA to have similar results.

If the 10-period SMA yields wildly better results than all other values for the SMA, that would imply that it caught a lucky trade, which has nothing to do with a 10-period being better than any other value.

Robustness in our parameters implies that the exact set of parameters is not critical in determining the strategy’s success. Rather, it is a robust idea that will have a higher probability of being repeatable.

When to Use Optimization

Dr Andrew Lo, at the 2022 CMTA Symposium in Washington, DC, stated his research had shown that optimization has no benefit when applied to model parameters, but does assist when applied to other portfolio inputs like position size and stops. We will explore these more in the Level III curriculum.

³ Fabozzi, Frank J., Sergio M. Focardi, and Petter N. Kolm. *Quantitative Equity Investing: Techniques and Strategies*. Hoboken, NJ: John Wiley & Sons, Inc., 2010.

⁴ Bailey, David H., Jonathan Borwein, Marcos López de Prado, and Qi Ji Jim Zhu. “Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest

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 Curriculum Update Log

 Sections

 Search



Curriculum
Update Log



Sections



Search

Topics

[Table of Contents](#)
[Confidence Levels](#)
[Notes](#)
[Bookmarks](#)

Strategy Update ▾

< > Strategy Update



12.2 Applying Quantitative Techniques

Strategy Update

As a final step in this unit, we will apply some of the concepts we discussed and see if we can improve our simple RRG idea. Quantitative analysis simply gives us a scorecard of our idea. Looking at the results, we should be making adjustments to see what improvements we can make.

The adjustments we made in this test:

1. Removed our ATR exit rule.
2. Added a trailing stop with percentage of 30%. This starts at 30% under our enter price and follows the trade up. As soon as the trade retraces 30%, the stop is hit.
3. Increased the position size from 4% to 5% (20 positions in the portfolio).

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Unit

12.2 Applying Quantitative Techniques

Measure	Original Value	New Value	Comment
<u>Portfolio Performance</u>			
Total Return	561%	719%	This is a good improvement on return.
Annual Return	11.5%	13.3%	
Annual Volatility	17.5%	20%	Less volatility is good.
CAGR	9.9%	11%	
Risk-Adjusted Returns	10.9	11%	
M^2	8.51	9.01	
Sharpe Ratio	0.5	0.53	
Sortino Ratio	2.85	2.48	
Treynor Ratio	10.5	10.1	
Calmar Ratio	0.25	0.23	
Maximum Drawdown	40%	49%	



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Sections



Search

(MDD)			
MDD Days	901	917	
Average R-Multiple Value	0.1	0.7	This is a lot better, as we would expect when we add a stop; our target is 1R. We still need to improve this, as there is not enough reward for the risk we are taking on.
SQN	4.4	3.7	This is a more realistic number, but highlights that we are not there yet.
<u>CAPM</u>			
Alpha	0.01	0.01	
Beta	0.84	1.04	
r^2	0.64	0.76	
<u>Benchmark Performance</u>			No changes, as we would expect.
Total Return	192%	192%	
CAGR	5.5%	5.5%	
Annual Return	4.4%	4.4%	
Annual Volatility	16.61%	16.61%	
Sharpe Ratio	0.25	0.25	
Maximum Drawdown (MDD)	56.8%	56.8%	
MDD Days	2,088	2,088	
<u>Trade Performance</u>			
Trades	907	275	Our trade stats are significantly improved. We are just



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Sections



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			not getting enough trades.
Gross Profit	\$10,520,248	\$9,412,361	
Gross Loss	-\$4,913,581	-\$2,214,062	The big difference is the reduction in gross loss.
Net Profit	\$5,606,667	\$7,198,299	
Profit Factor	2.14	4.25	This is a significant increase.
Average Profit	\$17,016	\$42,278	
Average Return	7.3%	29.7%	
Max Adverse Excursion	-24.3%	-21.6%	
Max Favorable Excursion	56.5%	42.9%	
Winning Trades	434	124	
Win Percent	47.9%	45.1%	Even fewer trades are winners.
Average Win	\$24,240	\$75,906	
Losing Trades	472	151	
Loss Percent	52%	54.9%	
Average Loss	-\$10,410	-\$14,663	
Trading Expectancy	\$6,197.76	\$26,183.62	
Max Profit	\$518,616	\$754,047	
Max Return	336.86%	679%	
Max Loss	-\$126,839	-\$100,544	
Min Return	-38.02%	-30%	With the stop set to 30%, this is what we would expect as the maximum percentage loss.

Summary

though there are fewer trades, we are doing a better job of utilizing the cash available. The real issue with this strategy is we are not getting enough opportunities. We will continue looking at how we can improve this in Level III.

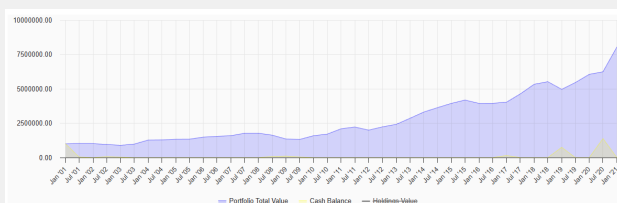


FIGURE 12.2.16 Updated equity plot

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